

Ensemble Quantum Computing Laboratory Prelab 1

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Problem 1

- (a) Free Induction Decay(FID),
pulse sequence:

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pulse(1, a90H, p1, d90H)
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Information found:

To begin with, by performing a FID, we can acquire the chemical shift of the nuclei in question.

In the case of spin-spin coupling, we can also acquire the information of the coupling constant. In the case of multiple excitation and acquisition, we can get the relaxation time.

- (b) Spin Echo,
pulse sequence:

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pulse(1, a90H, p1, d90H)
wait(w1)
pulse(1, a90H, p1, d180H)
wait(w1)
```

Information found:

By conducting spin echo, the effect of inhomogeneity of the magnetic field can be eliminated, allowing us to acquire the real T2.

I believe this experiment is designed to measure the T2 relaxation time. However, there are other information we can acquire as side product such as the T1, the chemical shift and the coupling strength.

- (c) Inversion Recovery,
pulse sequence:

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pulse(1, a90H, p1, d180H)
wait(w1)
pulse(1, a90H, p1, d90H)
```

Information found:

By conducting inversion recovery, we can acquire the T1 relaxation time, the time for spins

to recover to equilibrium.

Problem 2

- (a) y-axis: dispersion mode

By shifting the pulse from x-axis to y-axis, the phase of the pulse is changed by $\pi/2$ degree. Assuming the original signal is $S(t) = \exp(i\phi)\exp(i\Omega t)\exp(-\frac{t}{T_2})$ and its FT is F_x , then the signal corresponding to pulse from y-axis is $S(t) = \exp(\frac{\pi}{2})\exp(i\phi)\exp(i\Omega t)\exp(-\frac{t}{T_2})$, by the linearity of FT, the FT of the signal is $F_y = \exp(\frac{\pi}{2})F_x$. Which is multiplying the original FT by a constant i , making the imaginary part of the original frequency domain (in dispersion mode because the real part is in absorption mode) the conjugation of current real part.

- (b) -x-axis: absorption mode, inverted

The time and frequency domain of the signal is now multiplied by -1 (phase difference π), therefore the signal is inverted in time domain and frequency domain.

- (c) 270 about x-axis: absorption mode, inverted

This is essentially the same as the previous one, the signal is inverted in time domain and frequency domain.

Problem 3

- (a) lab frame Hamiltonian and rot. fram. Hamiltonian:

A generalized enough magnetic field in this case is: $B_0\hat{z} + B_1(\cos(\omega t + \phi) + \sin(\omega t + \phi))$, the Hamiltonian of a magnetic moment in this field is $H = -\vec{\mu} \cdot \vec{B} = -\gamma\vec{S} \cdot \vec{B}$, where γ is the gyromagnetic ratio.

lab frame Hamiltonian:

$$H_{lab} = \frac{\hbar\omega_0}{2}\hat{\sigma}_z + \frac{\hbar\omega_1}{2}(\cos(\omega(t) + \phi)\hat{\sigma}_x + \sin(\omega(t) + \phi)\hat{\sigma}_y).$$

lab frame unitary propagator:

This Hamiltonian is time dependent therefore difficult to solve its unitary propagator, the unitary propagator is solved by transforming the Hamiltonian to rotating frame, calculate the propagator and transform it back to lab frame.

$$U_{lab}(t) = \exp[\frac{i}{2}(\omega t + \phi)]\exp[-\frac{i}{2}\Omega_R\hat{n} \cdot \sigma t]$$

$$\text{where } \Omega_R = \sqrt{(\omega_0 - \omega)^2 + \omega_1^2}, \text{ and } \hat{n} = \frac{1}{\Omega_R}(\omega_1\cos\phi, \omega_1\sin\phi, \omega_0 - \omega)$$

rotating frame Hamiltonian:

After making the Rotating Wave Approximation(RWA), and assume the RF field is on resonance, we get $H_{rot.frame.}(t) = \frac{\hbar}{2}(\omega_0 - \omega)\sigma_z + \frac{\hbar\omega_1}{2}(\cos(\phi)\sigma_x + \sin(\phi)\sigma_y)$

rotating frame unitary proporgator:

$$\cos(\frac{\Omega_R t}{2})\mathbb{I} - i\sin(\frac{\Omega_R t}{2})\hat{n} \cdot \sigma$$

(b) evolution of the state under ther given magnetic field:

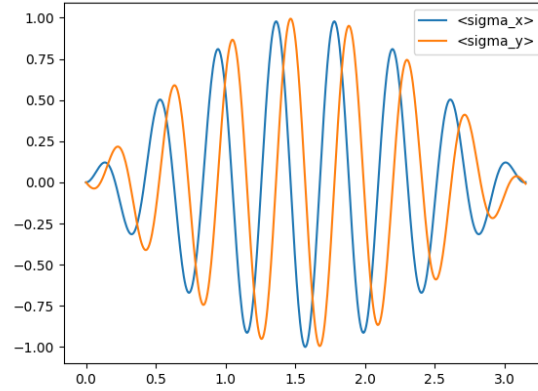


Figure 1: Observation of the state under the given magnetic field, in x and y basis

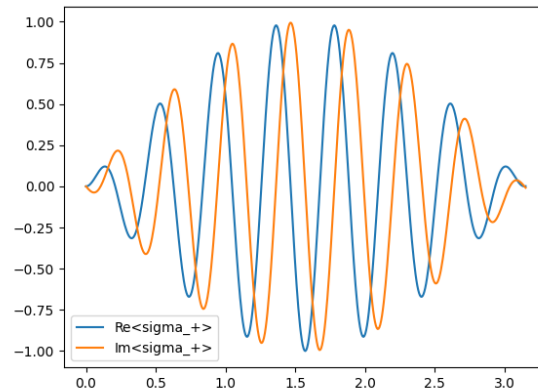


Figure 2: Observation of the state under the given magnetic field, in $\sigma+$ and $\sigma-$ basis

(c) evaluate signal in x , y and $\sigma+$ basis:

see above

(d) the bloch sphere representation

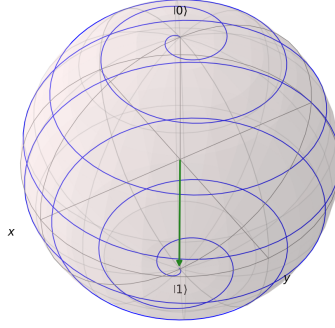


Figure 3: Observation of the state under the given magnetic field, presented in bloch sphere, the blue line is the trajectory of the state vector

Problem 4

$$\begin{aligned}
 & \text{one-sided FT for } e^{i(\omega_0 t - \phi)} e^{-\frac{t}{T_2}} \text{ is } \int_0^{+\infty} e^{-i\omega t} e^{i(\omega t - \phi)} e^{-\frac{t}{T_2}} dt \\
 &= e^{-i\phi} \int_0^{+\infty} e^{-\frac{t}{T_2}} e^{i(\omega - \omega_0)t} dt \\
 &= e^{-i\phi} \int_0^{+\infty} e^{[-\frac{1}{T_2} + i(\omega - \omega_0)]t} dt \\
 &= e^{-i\phi} \frac{1}{-\frac{1}{T_2} + i(\omega - \omega_0)}
 \end{aligned}$$

notice if $\Gamma = -\frac{1}{T_2}$ and $\alpha = \frac{e^{-i\phi}}{-1\frac{1}{T_2}}$, then $g(\omega) = \frac{\alpha\Gamma}{i(\omega - \omega_0) + \Gamma}$, the standard parameterization of lorentzian function.

Problem 5

Since the typical linewidth in question is 25Hz, the maximum acceptable change in frequency is 2.5Hz. The Homogeneity in question is 1per10⁸, for Lamor frequency of 180MHz, meaning the change in given magnetic field is 1.8Hz, which is acceptable. Meaning the magnetic field is sufficiently homogeneous.

Problem 6

A “two bit ADC” is a device that can convert the amplitude of a analog signal to a digital signal with 2 bits. The output of the ADC is a binary number with 2 bits.

The reason why it is often generally desirable to have a higher bit ADC is that the higher the bit, the higher the resolution of the digital signal is. The higher the resolution, the more accurate the digital signal is to the analog signal.

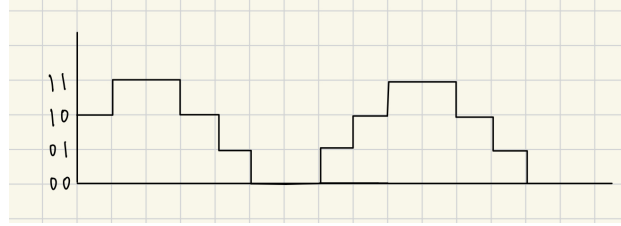


Figure 4: A possible digital signal for sin signal after a two bit ADC

Problem 7

For a spectrometer operating at 800MHz, in order to cover 15ppm, the spectral range of the spectrometer should be $800\text{MHz} \times 15\text{ppm} = 12\text{kHz}$.

In the case of quadrature detection, the dwell time should be $1/12\text{kHz} = 83.3\mu\text{s}$.

I have some doubts about my interpretation of the question, I am not certain if it means the total range is 15ppm or the spectrometer should cover 15ppm on each side of the center frequency. I am assuming the prior assumption, in the case of the latter, we simply need to double the spectral range and half the dwell time.